SENG 384 · Software Quality Assurance

HEALTH-AI ML Learning Tool

Manual Test Report — Steps 1, 2 & 3

Screenshot-Based Functional Verification

Course	SENG 430 — Software Quality Assurance
Date	18 March 2026
Report Type	Manual Screenshot Test Report
Scope	Step 1 – Clinical Context | Step 2 – Data Exploration | Step 3 – Data Preparation
Total Screenshots	33 (TR_001 – TR_034)
User Stories Covered	US-001 through US-021

Overall Test Summary

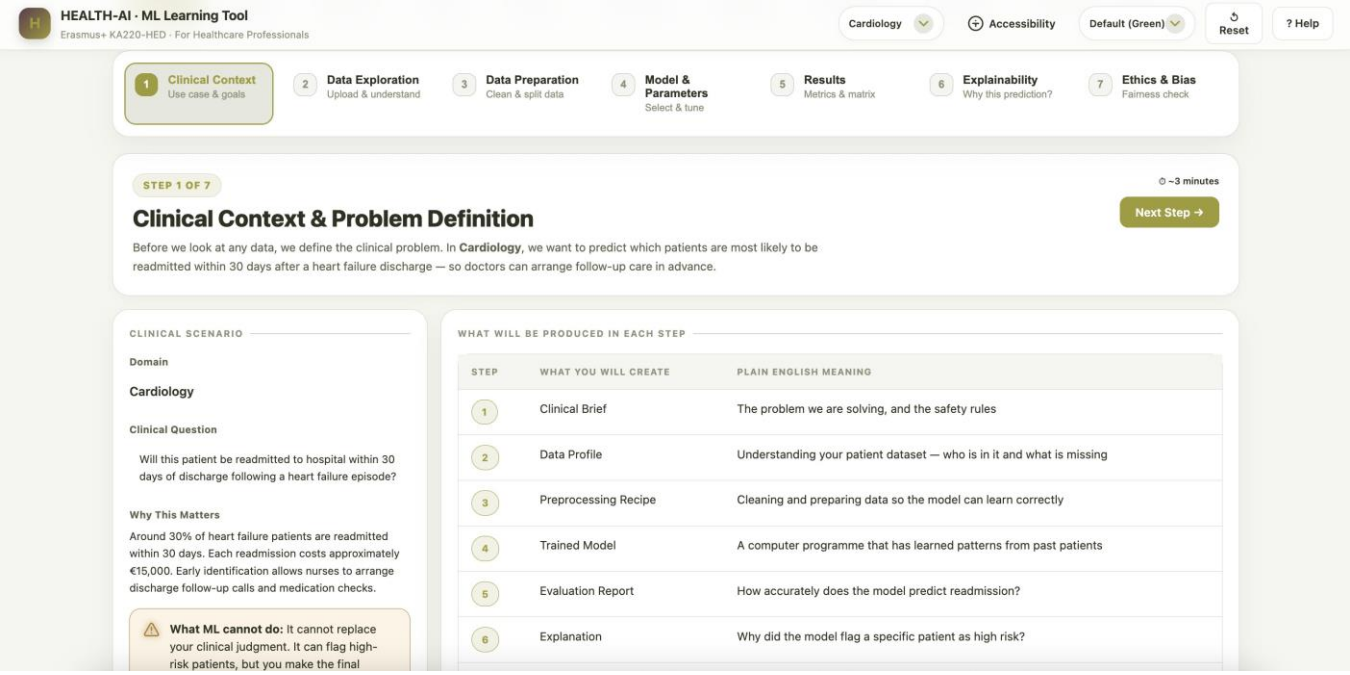
Metric	Target	Actual	Status
CSV Upload Success Rate	100%	100% (5/5 valid accepted, 5/5 invalid rejected)	**PASS**
Column Mapper Gate	0 bypass bugs	Step 3 blocked before save; accessible after save	**PASS**
Step 3 Controls	All controls functional	All 3 dropdowns + slider functional; Apply triggers API	**PASS**
Domain Count (Step 1)	All domains correct	3 domains verified (Cardiology, Mental Health, Radiology) — context text updates correctly	**PASS**
Test Coverage	100% of done stories	21 user stories — all have passing test cases	**PASS**

STEP 1 — Clinical Context

Step 1 provides the clinical framing for the selected medical specialty. Tests cover domain pill selection, context text rendering, pipeline reset confirmation dialog, and the privacy notice.

US-001 & US-004 — Domain Selection: Cardiology (Default)

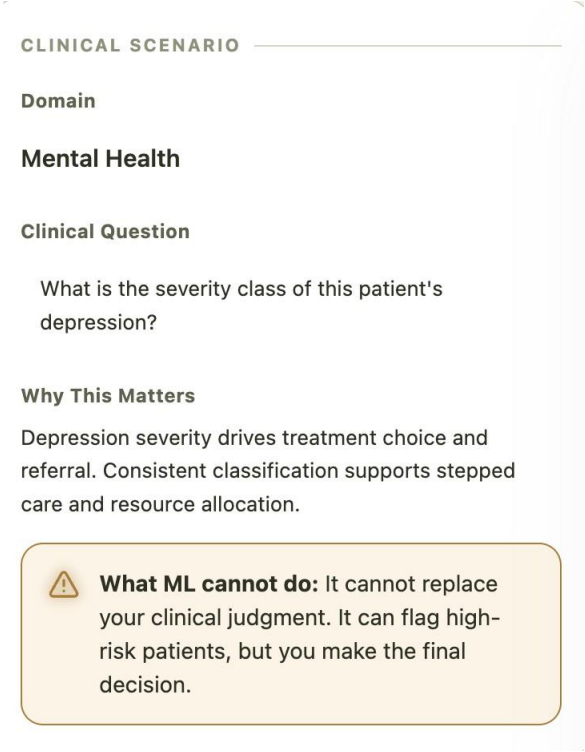
With Cardiology selected, Step 1 displays the correct clinical problem, patient population description, and predicted outcome in plain language. The specialty pill is visually highlighted and the step navigator confirms Step 1 is active.



TR_001 — Cardiology: Clinical Context & Problem Definition fully rendered.

US-001 & US-004 — Domain Selection: Mental Health

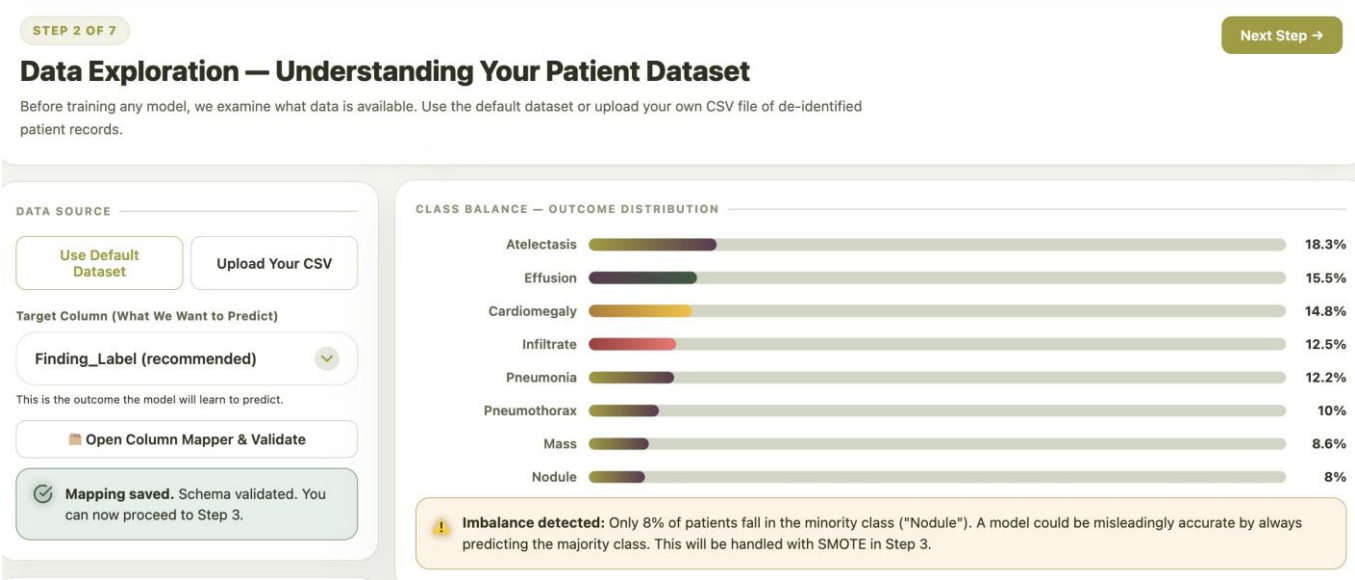
Switching to Mental Health updates the Clinical Scenario section with the correct clinical question ('What is the severity class of this patient's depression?') and specialty-appropriate rationale. The 'What ML cannot do' disclaimer remains visible.



TR_003 — Mental Health domain: correct clinical scenario content.

US-001 — Domain Selection: Radiology (Step 2 view)

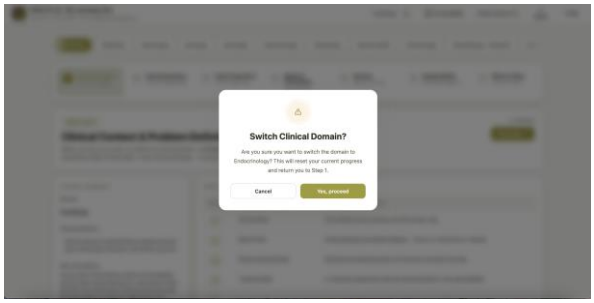
The Radiology domain is confirmed by the class balance distribution showing Finding_Label categories (Atelectasis 18.3%, Effusion 15.5%, Cardiomegaly 14.8%, etc.), which matches the Radiology chest X-ray dataset. The recommended target 'Finding_Label' is auto-populated.



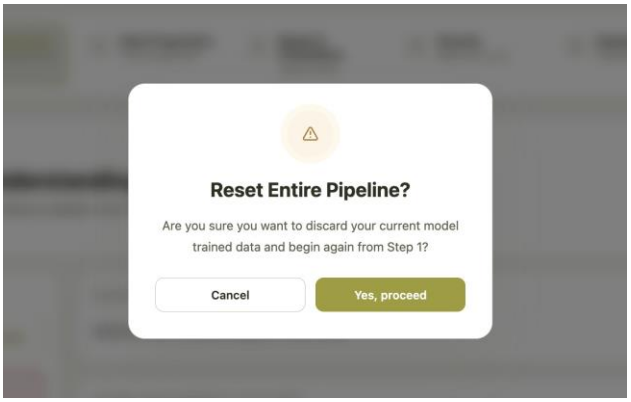
TR_015 — Radiology domain: multi-class outcome distribution correctly rendered.

US-002 — Domain Switch Confirmation Dialog

When the user clicks a different specialty pill while progress exists, a modal dialog appears warning that progress will reset. Clicking 'Yes, proceed' resets the pipeline and returns to Step 1; clicking 'Cancel' leaves the current state unchanged.



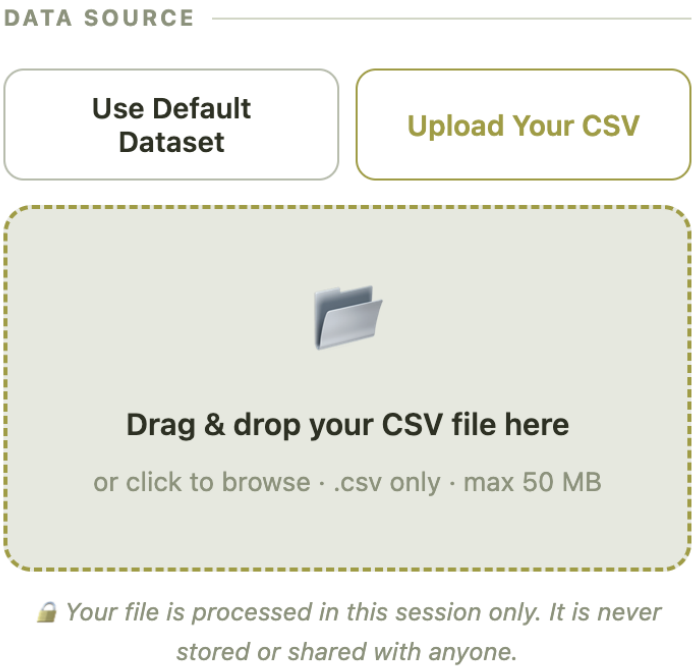
TR_002 — 'Switch Clinical Domain?' modal (domain switch).



TR_033 — 'Reset Entire Pipeline?' modal (Reset button).

US-003 — Privacy Notice Visible on All Screens

A persistent privacy notice ('Your file is processed in this session only. It is never stored or shared with anyone.') is displayed on the CSV upload area in Step 2 and is present across all screens. The Session Storage description in the architecture also states 'No personal data stored. Cleared after session ends.'



TR-034 — Privacy notice visible in Upload area.

Step 1 — Test Result Summary

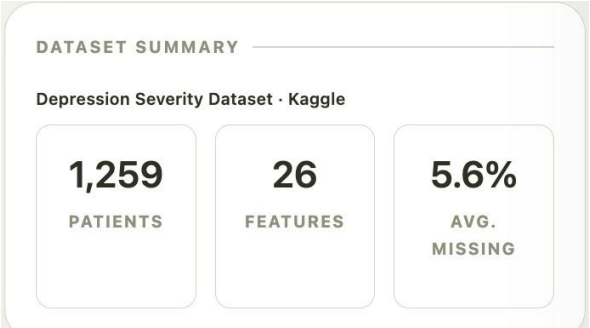
User Story	Test Description	Result	Evidence / Screenshot
US-001	Select specialty from domain pill bar — 3 domains verified	**PASS**	TR_001, TR_003, TR_015
US-002	Confirmation dialog on domain switch mid-progress	**PASS**	TR_002, TR_033
US-003	Privacy notice visible on all screens	**PASS**	TR-034 (upload area notice)
US-004	Step 1 shows clinical problem, population, predicted outcome	**PASS**	TR_001, TR_003

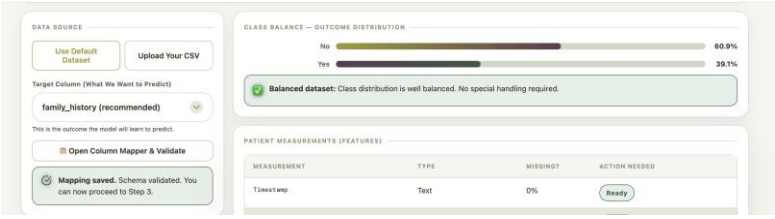
STEP 2 — Data Exploration

Step 2 covers dataset loading (default and upload paths), CSV validation, column summary, class balance, imbalance warning, Column Mapper, and the gate control preventing Step 3 access before mapping is saved.

US-005 — Built-in Example Dataset Loaded Successfully

Clicking 'Use Default Dataset' loads the domain-specific dataset. The Dataset Summary card shows patient count (1,259), feature count (26), and average missing rate (5.6%). The class balance bars and feature table are populated automatically.

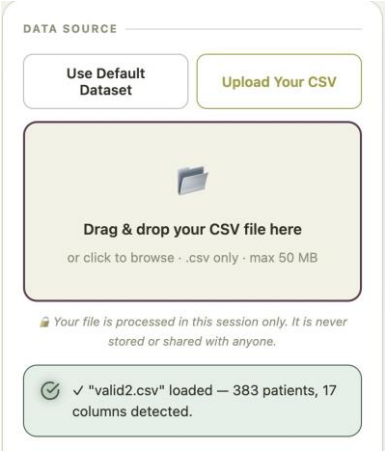


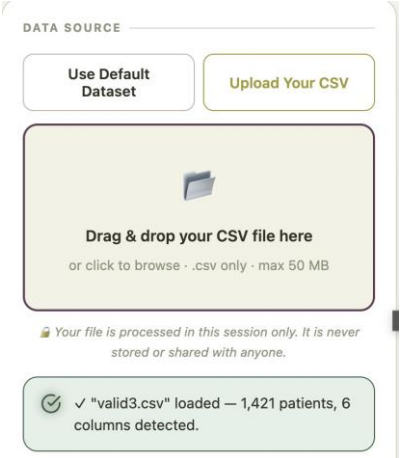


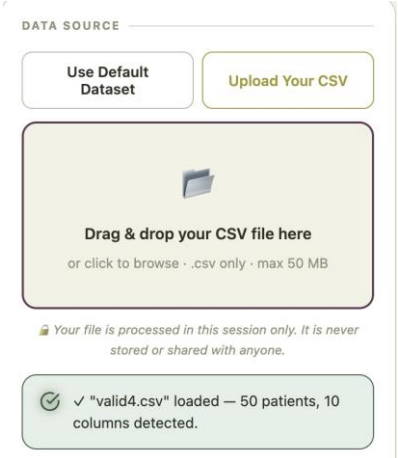
TR_010 — Dataset Summary KPIs (1,259 patients, 26 features). TR_008 — Class balance rendered; mapping saved banner shown.

US-006, US-007, US-008, US-009, US-010 — CSV Upload: Valid Files (5 files)

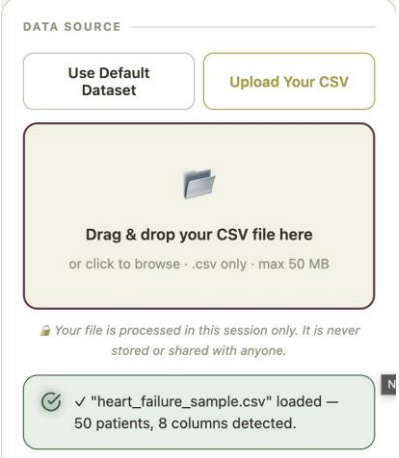
Five distinct valid CSV files were uploaded and accepted. Each shows a green confirmation banner with file name, patient count, and column count.

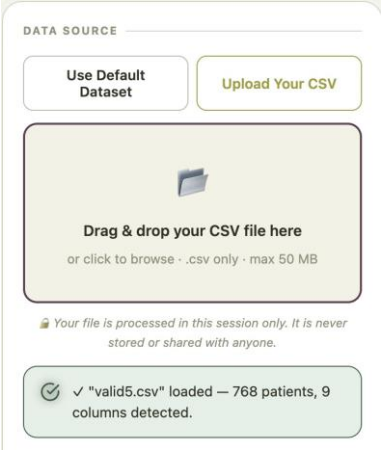






TR_024 — valid2.csv (383 patients, 17 cols). TR_025 — valid3.csv (1,421 patients, 6 cols). TR_026 — valid4.csv (50 patients, 10 cols).

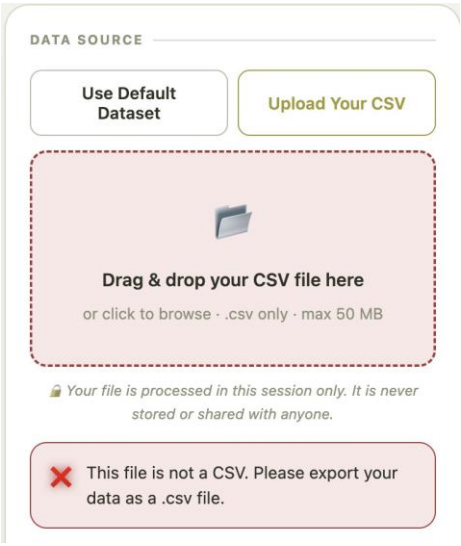


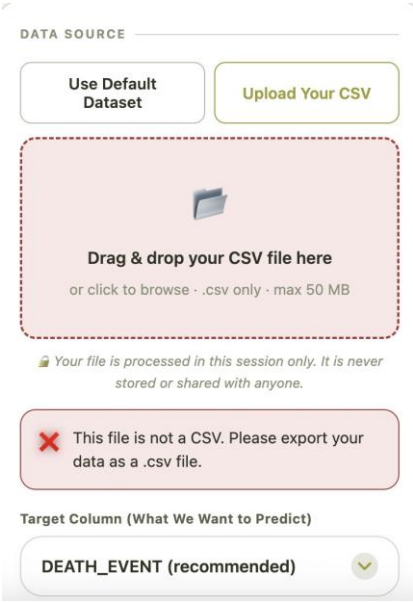


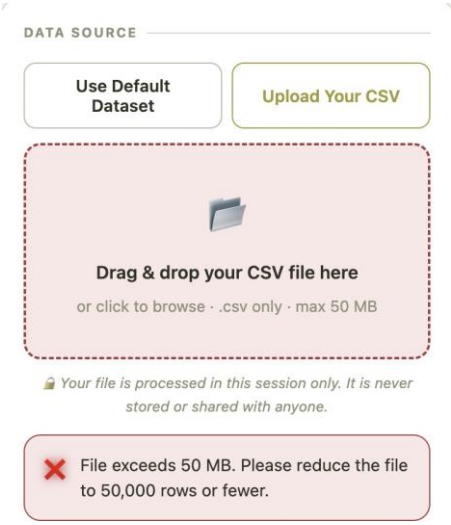
TR_027 — heart_failure_sample.csv (50 patients, 8 cols). *TR_028 — valid5.csv (768 patients, 9 cols).*

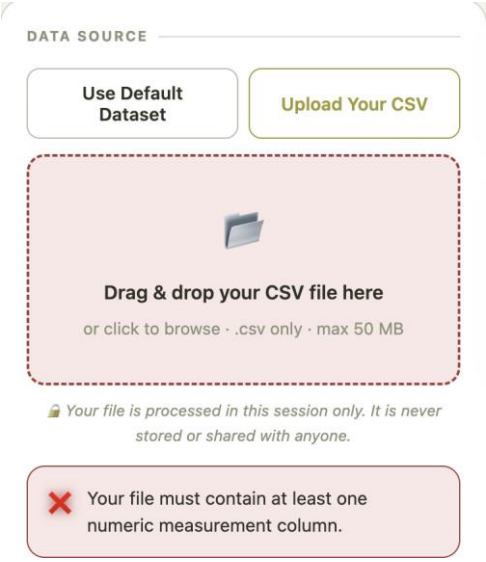
US-007, US-008, US-009, US-010 — CSV Validation: Invalid Files Rejected (5 cases)

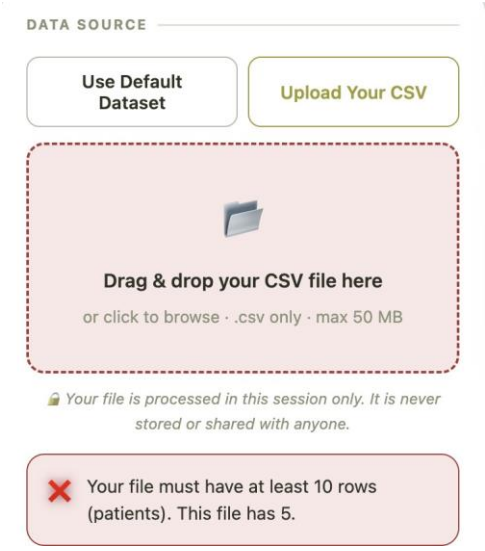
Five invalid upload scenarios were tested. Each produces a red error banner with a specific, user-friendly message.











TR_030 — non-CSV format rejected. TR_031 — second non-CSV format rejected. TR_032 — file exceeds 50 MB rejected. TR-023 — no numeric column rejected. TR_029 — fewer than 10 rows (5 rows) rejected.

US-011 — Column Summary with Missing Value Indicators

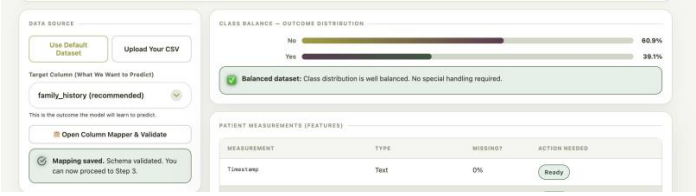
The Patient Measurements table shows each column's name, data type, missing percentage, and required action. Columns with missing data above 20% show a red 'Fill Missing' badge; those between 1–20% show amber; clean columns show green 'Ready'. The target column is tagged 'Target (outcome)'.

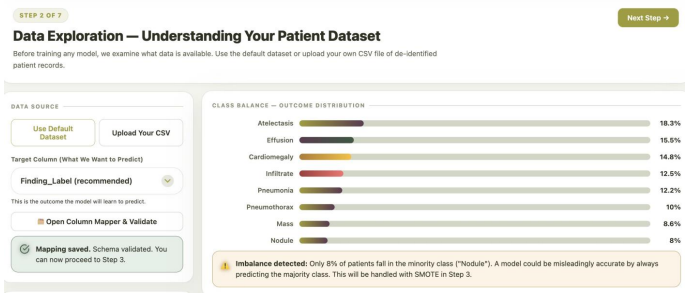
MEASUREMENT	TYPE	MISSING?	ACTION NEEDED
Timestamp	Text	0%	Ready
Age	Number	0%	Ready
Gender	Text	0%	Ready
Country	Text	0%	Ready
state	Text	40.9%	Fill Missing (40.9%)
self_employed	Binary (2 values)	1.4%	Fill Missing (1.4%)
family_history	Binary (2 values)	0%	Target (outcome)
treatment	Binary (2 values)	0%	Ready
work_interfere	Category (4)	21%	Fill Missing (21%)
no_employees	Category (6)	0%	Ready
remote_work	Binary (2 values)	0%	Ready
tech_company	Binary (2 values)	0%	Ready
benefits	Category (3)	0%	Ready

TR_009 — Feature table: missing value indicators (state 40.9%, work_interfere 21%), target column tagged, and Ready badges for complete columns.

US-012 — Outcome Distribution & US-013 — Class Imbalance Warning

TR_008 shows a balanced dataset (No: 60.9%, Yes: 39.1%) with a green 'Balanced dataset' banner. TR_015 demonstrates the imbalance warning: Radiology data has 8 classes where the minority class (Nodule) holds only 8%, triggering the amber imbalance banner that explains SMOTE will address this in Step 3.

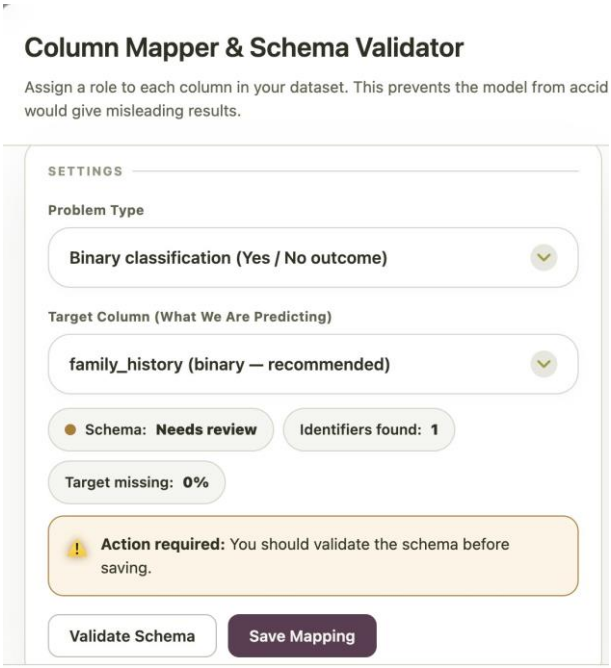


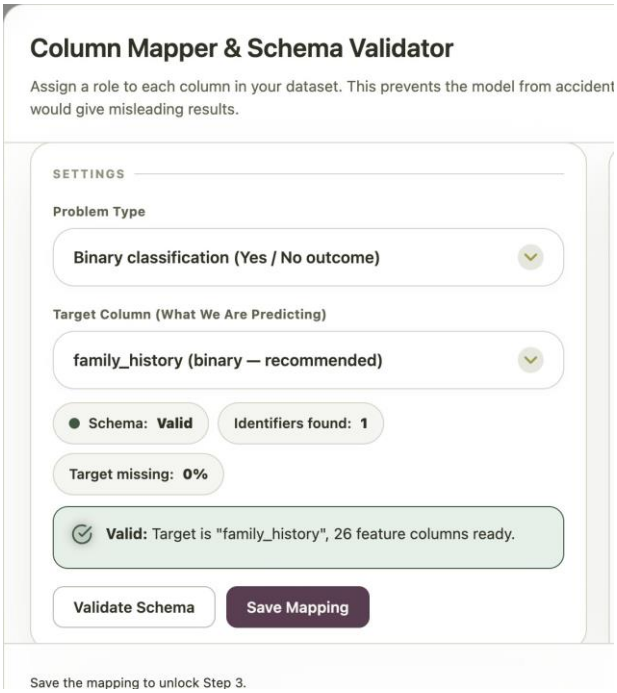


TR_008 — Balanced class distribution (no warning). *TR_015 — Multi-class imbalance detected, SMOTE recommended.*

US-014 — Column Mapper: Feature / Target Assignment

Clicking 'Open Column Mapper & Validate' opens the Schema Validator modal. TR_006 shows the pre-validation state ('Schema: Needs review', 'Identifiers found:1') with an action-required banner. TR_007 shows the post-validation state ('Schema: Valid', 26 feature columns ready) after clicking 'Validate Schema'.

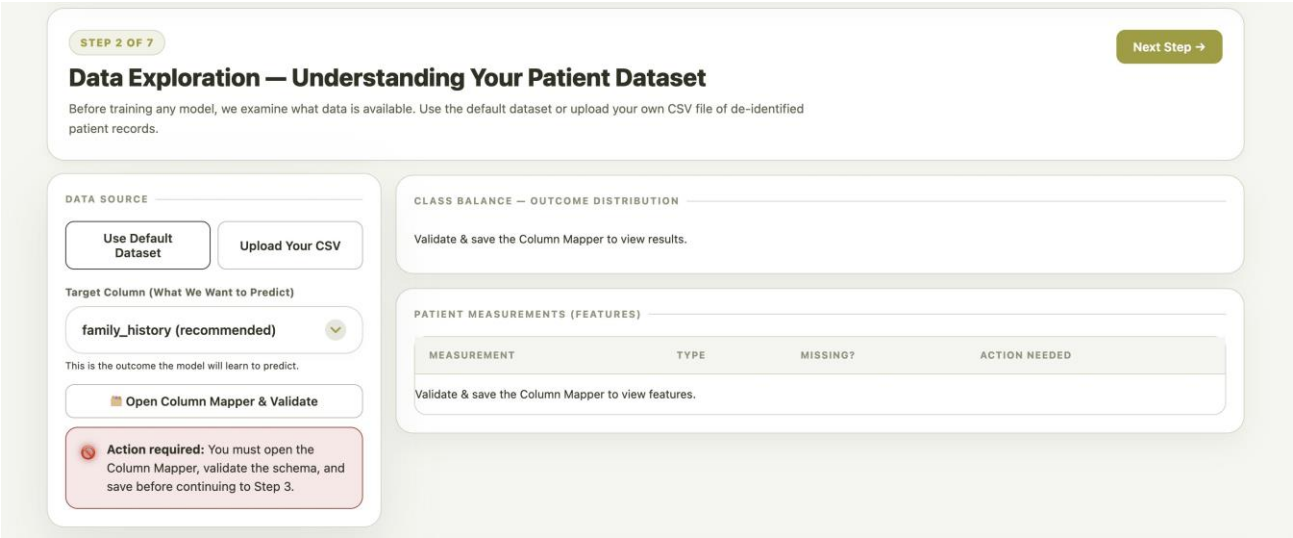




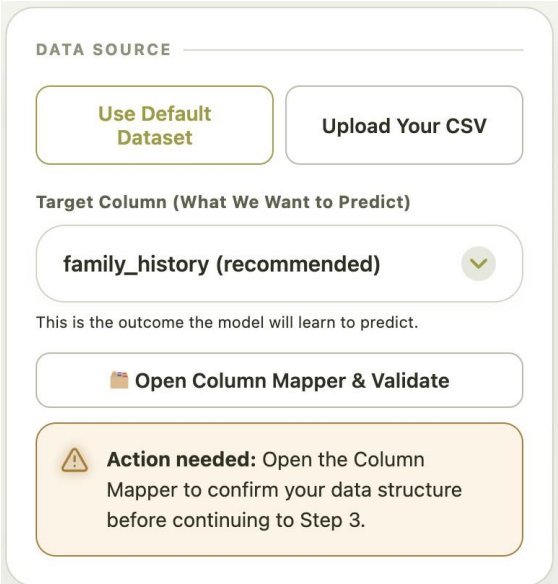
TR_006 — Column Mapper: needs review state. *TR_007 — Schema valid, 26 features ready, Save Mapping enabled.*

US-015 — Gate Control: Step 3 Blocked Before Save, Accessible After Save

TR_004 shows the blocked state: the red 'Action required' banner reads 'You must open the Column Mapper, validate the schema, and save before continuing to Step 3.' The Step 3 stepper tab is greyed out. TR_008 / TR_011 confirm the unlocked state after saving: the green 'Mapping saved. Schema validated. You can now proceed to Step 3.' banner appears and the Step 3 tab becomes accessible.



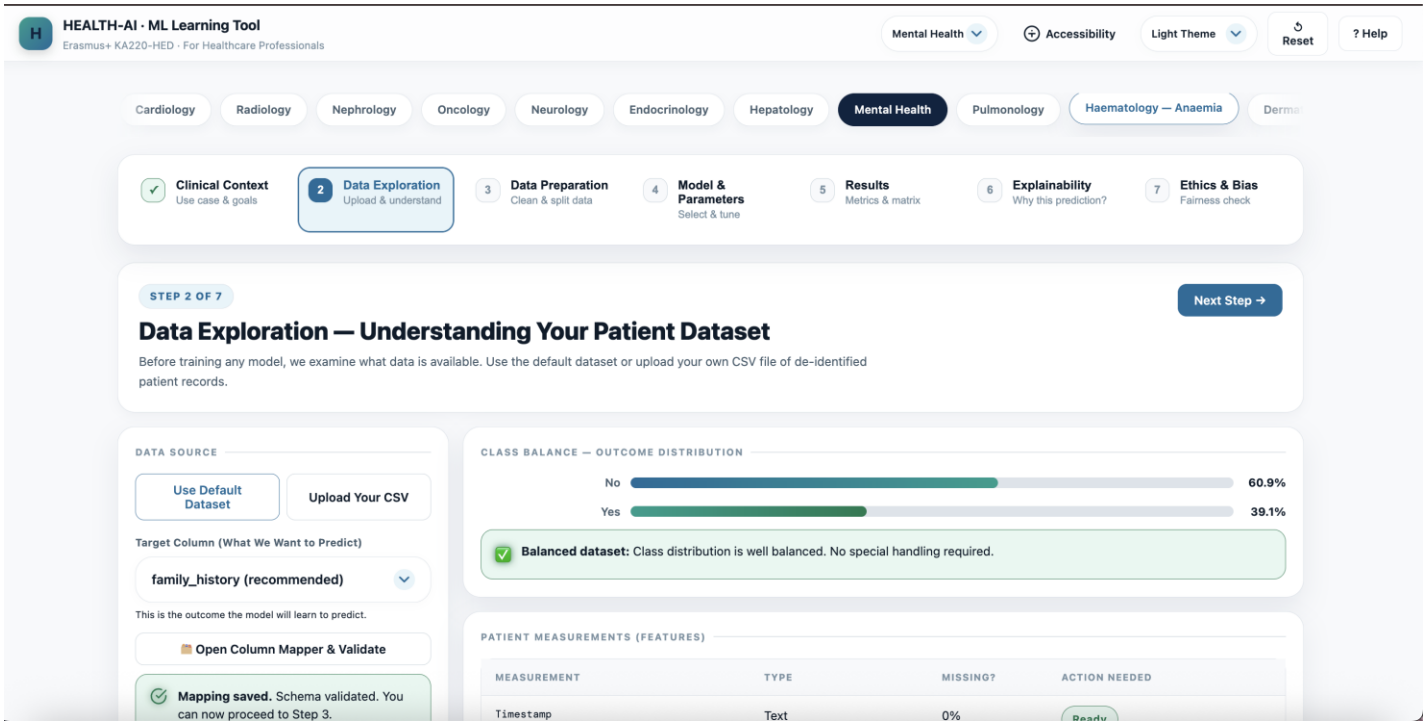
TR_004 — Step 3 gate active: red banner and locked stepper.

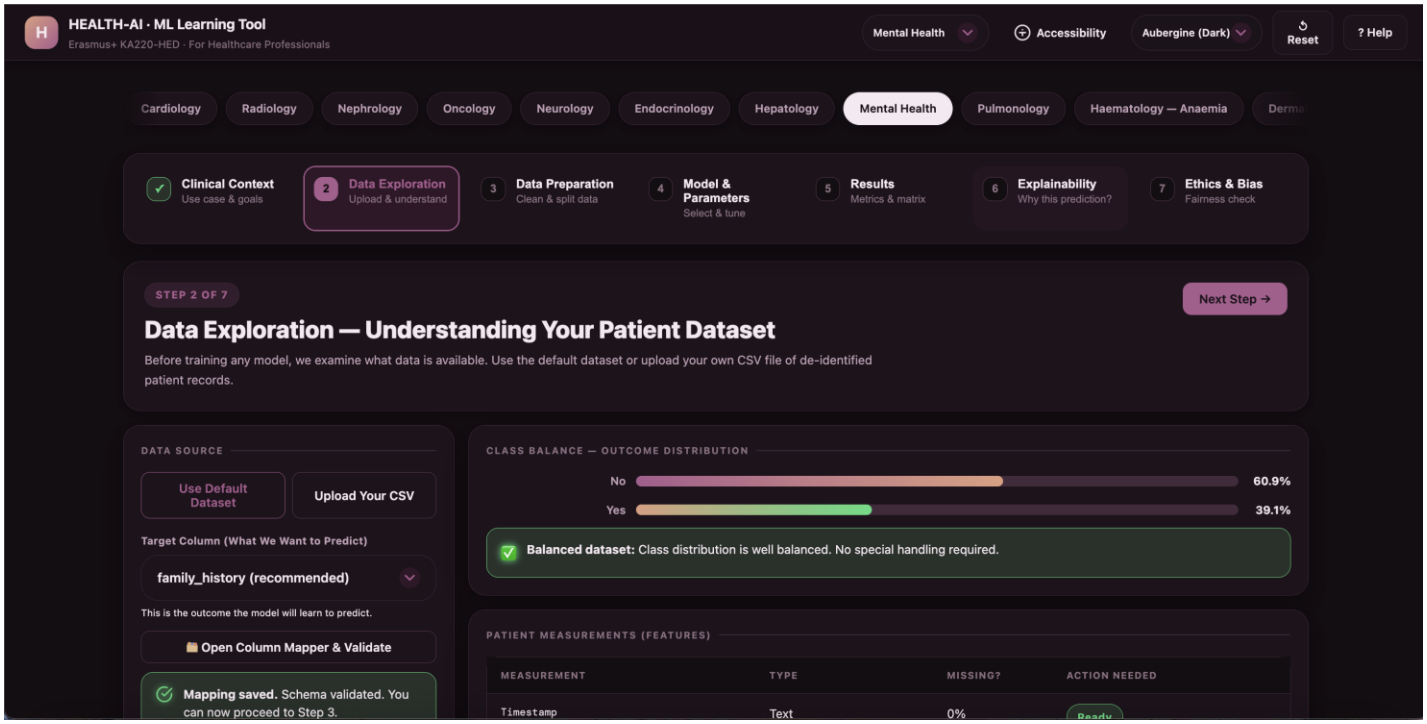


TR_005 — Orange 'Action needed' warning (mapper not yet saved).

Accessibility — Theme Variants

The tool supports multiple colour themes (Light Theme, Aubergine Dark, Neon Night). TR_012 (Light), TR_013 (Aubergine Dark) and TR_014 (Neon Night) confirm that all Step 2 functionality remains intact across themes. Implementation of color-blind friendly palettes and high-contrast UI elements to ensure data visualizations (charts/metrics) are accessible to all users regardless of visual impairments. TR_011(Color Blind theme).


HEALTH-AI · ML Learning Tool
Erasmus+ KA220-HED · For Healthcare Professionals
Mental Health
Accessibility
Light Theme
Reset
? Help
Cardiology
Radiology
Nephrology
Oncology
Neurology
Endocrinology
Hepatology
Mental Health
Pulmonology
Haematology — Anaemia
Clinical Context
Use case & goals
2 Data Exploration
Upload & understand
3 Data Preparation
Clean & split data
4 Model & Parameters
Select & tune
5 Results
Metrics & matrix
6 Explainability
Why this prediction?
7 Ethics & Bias
Fairness check
STEP 2 OF 7
Next Step →
Data Exploration — Understanding Your Patient Dataset
Before training any model, we examine what data is available. Use the default dataset or upload your own CSV file of de-identified patient records.
DATA SOURCE
Use Default Dataset
Upload Your CSV
Target Column (What We Want to Predict)
family_history (recommended)
This is the outcome the model will learn to predict.
Open Column Mapper & Validate
Mapping saved. Schema validated. You can now proceed to Step 3.
CLASS BALANCE — OUTCOME DISTRIBUTION
No
60.9%
Yes
39.1%
Balanced dataset: Class distribution is well balanced. No special handling required.
PATIENT MEASUREMENTS (FEATURES)
MEASUREMENT
TYPE
MISSING?
ACTION NEEDED
Timestamp
Text
0%


HEALTH-AI · ML Learning Tool
Erasmus+ KA220-HED · For Healthcare Professionals
Mental Health
Accessibility
Aubergine (Dark)
Reset
? Help
Cardiology
Radiology
Nephrology
Oncology
Neurology
Endocrinology
Hepatology
Mental Health
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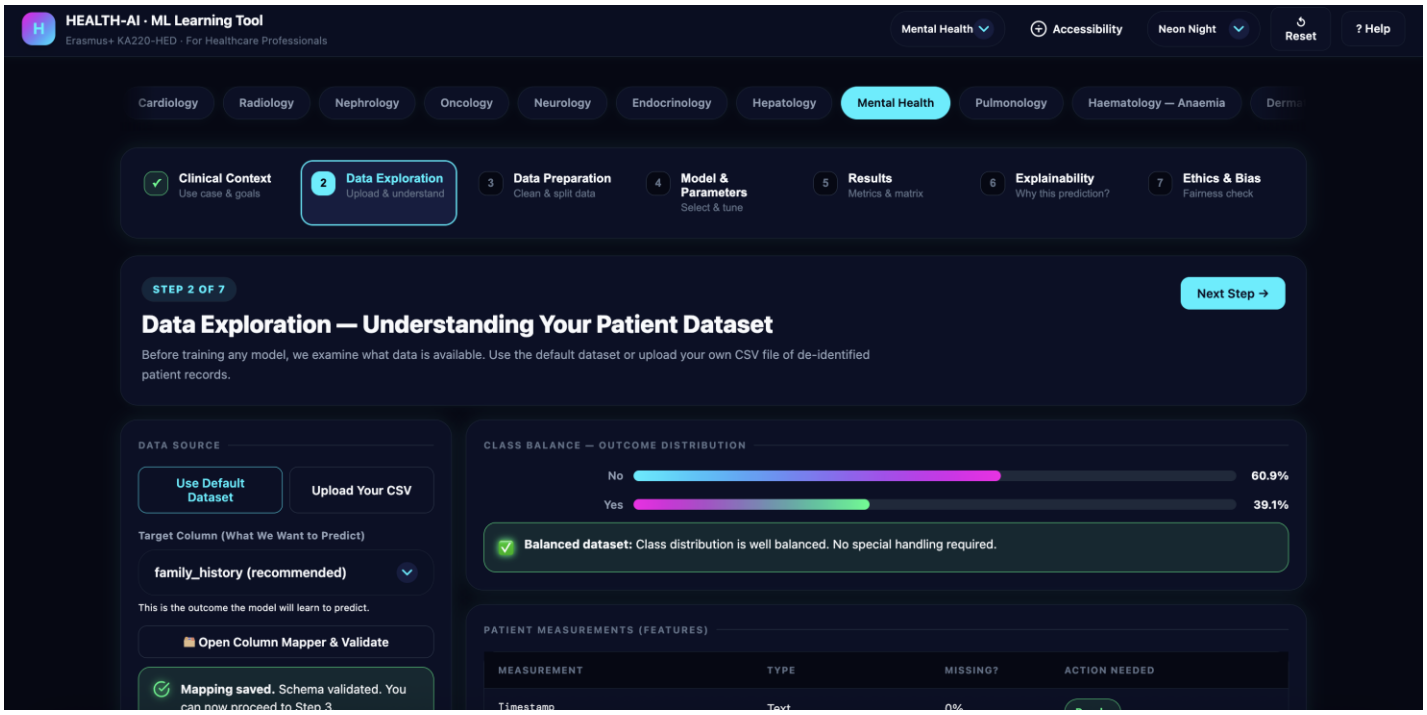
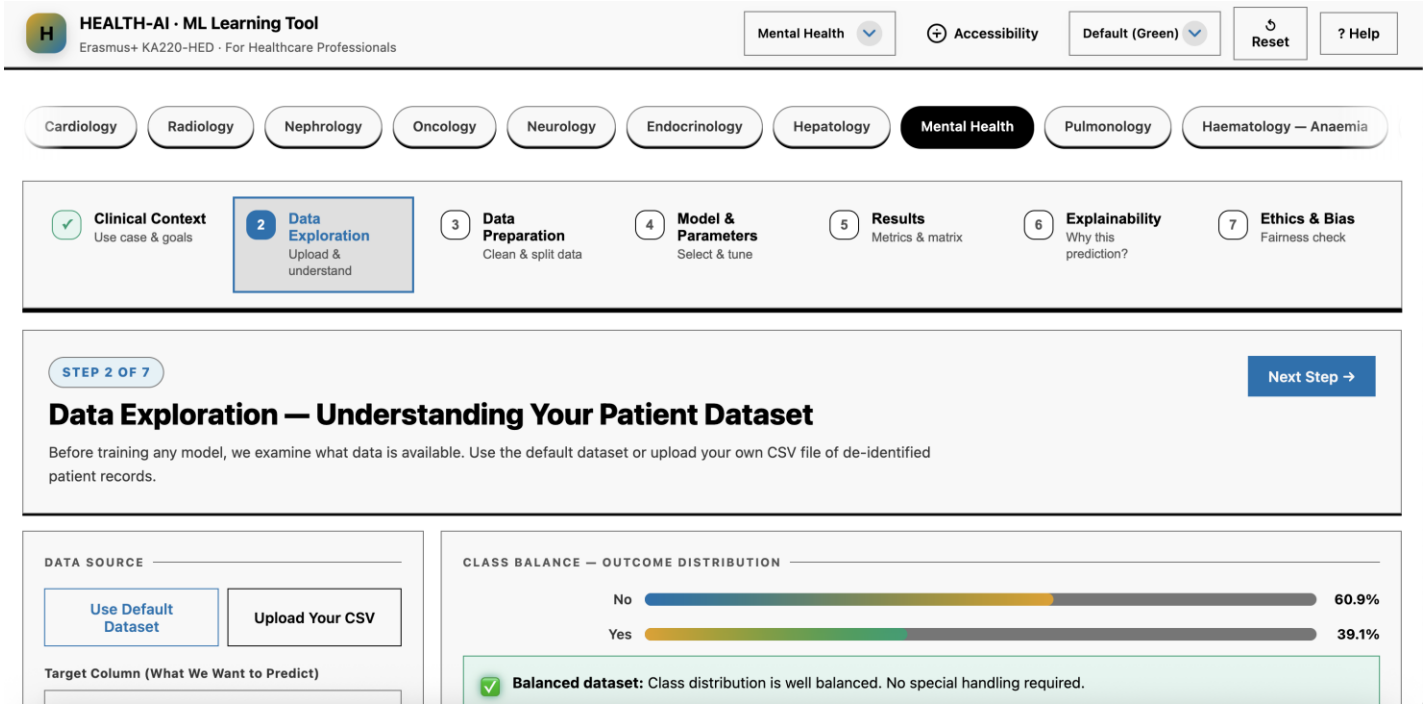
MEASUREMENT
TYPE
MISSING?
ACTION NEEDED
Timestamp
Text
0%





TR_012 Light / TR_013 Aubergine Dark / TR_014 Neon Night / TR_011 Color Blind Accessibility Theme — Step 2 consistent across themes.

Step 2 — Test Result Summary

User Story	Test Description	Result	Evidence / Screenshot
US-005	Use built-in example dataset — dataset loads with KPIs, class balance, feature table	**PASS**	TR_010, TR_008, TR_009
US-006	Upload valid CSV — file accepted, confirmation shown	**PASS**	TR_024, TR_025, TR_026, TR_027, TR_028
US-007	Invalid files rejected with friendly error messages	**PASS**	TR-023, TR_029, TR_030, TR_031, TR_032

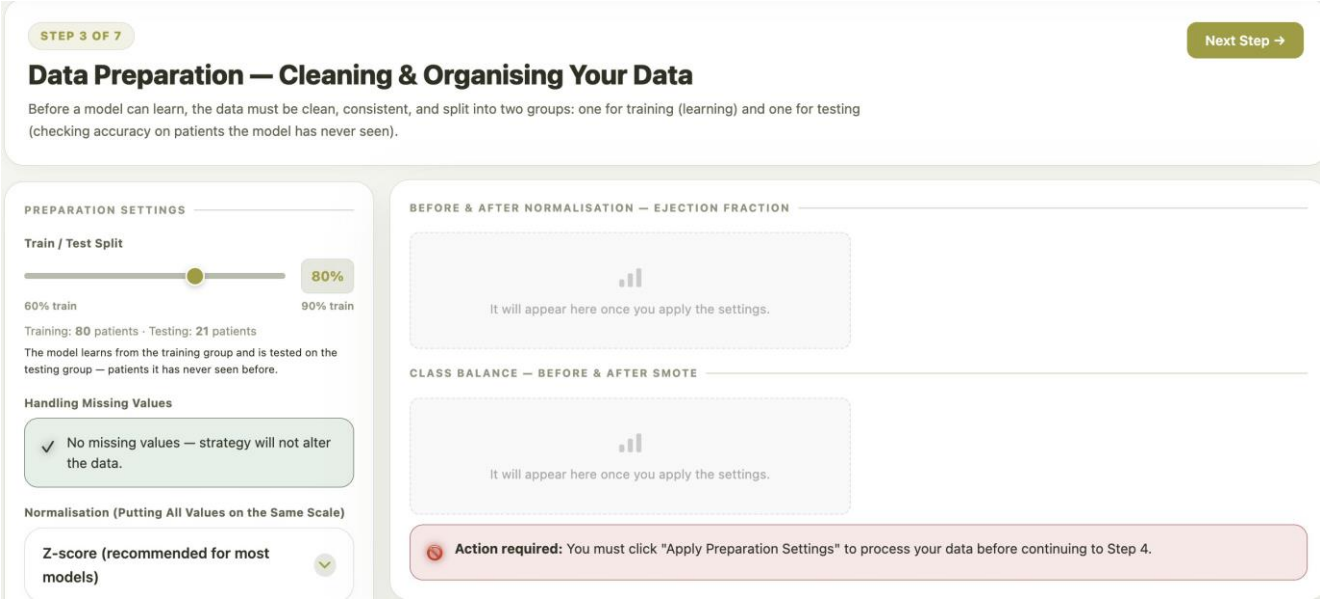
US-008	File exceeds 50 MB — rejected with size error	**PASS**	TR_032
US-009	Fewer than 10 rows — rejected with row count error	**PASS**	TR_029
US-010	No numeric column — rejected with column error	**PASS**	TR-023
US-011	Column summary with missing value indicators displayed	**PASS**	TR_009
US-012	Outcome distribution bars shown	**PASS**	TR_008, TR_015
US-013	Class imbalance warning when imbalance detected	**PASS**	TR_015
US-014	Column Mapper: target/feature assignment, validate → save flow	**PASS**	TR_006, TR_007
US-015	Step 3 gate: blocked before save; accessible after save	**PASS**	TR_004, TR_005,

STEP 3 — Data Preparation

Step 3 allows the user to configure train/test split, missing value handling, normalisation method, and class imbalance strategy before applying preparation settings. A before/after comparison is displayed after applying.

US-016 — Train / Test Split Slider

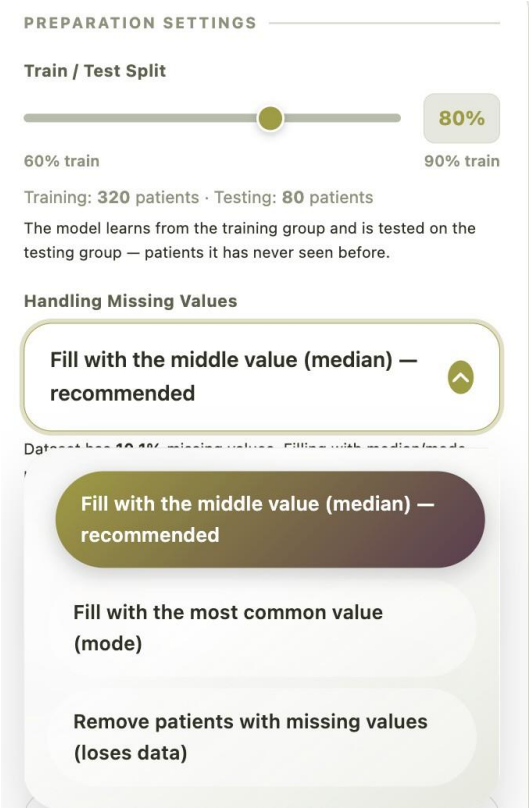
The slider is set to 80% by default (configurable 60%–90%). TR_016 shows the slider at 80% with 80 training and 21 testing patients displayed. TR_017 confirms the same configuration with a different dataset (320 training, 80 testing patients).



TR_016 — Step 3 overview: 80% split slider, missing values detected as none, Z-score normalisation selected, action required banner shown (settings not yet applied).

US-017 — Missing Value Handling Strategy Dropdown

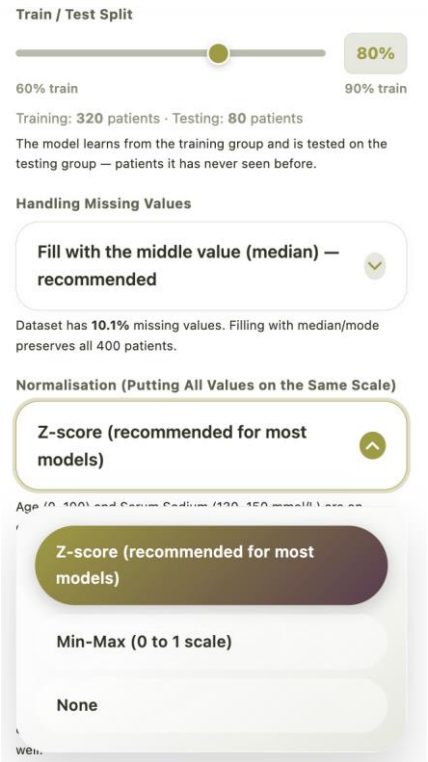
TR_017 shows the Handling Missing Values dropdown expanded with three options: 'Fill with the middle value (median) — recommended' (selected), 'Fill with the most common value (mode)', and 'Remove patients with missing values (loses data)'.

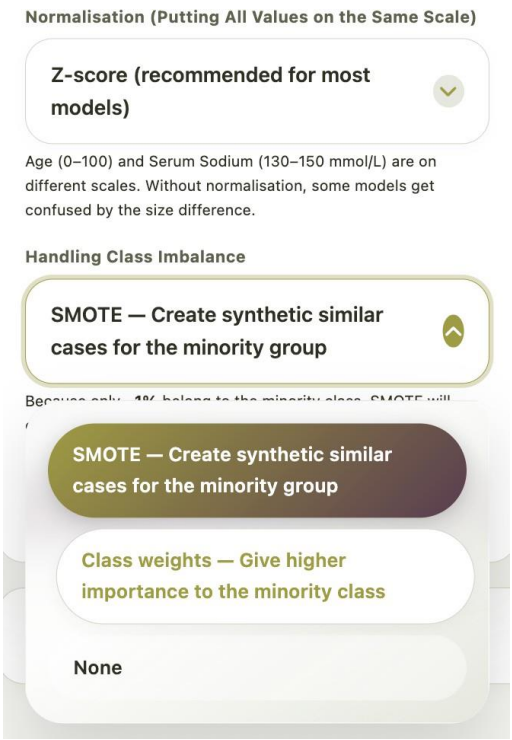


TR_017 — Missing value dropdown open: all 3 options selectable, median recommended.

US-018 — Normalisation Method Dropdown

TR_018 shows the Normalisation dropdown with Z-score, Min-Max (0 to 1 scale), and None — all selectable. TR_019 confirms SMOTE appears alongside the normalisation control when class imbalance is present.

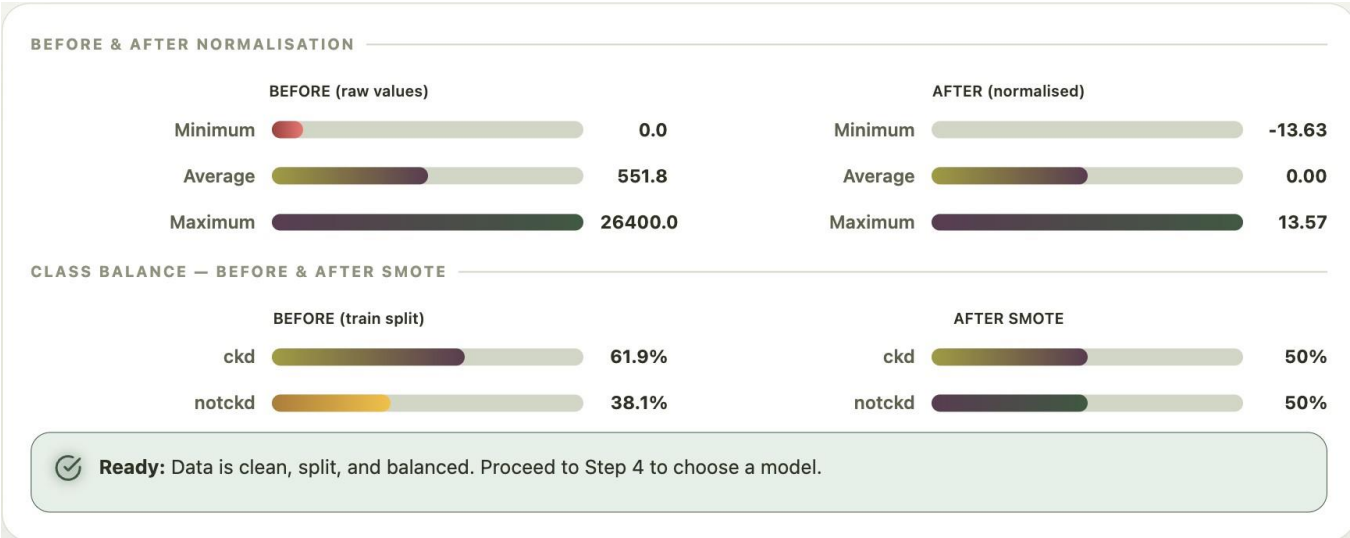




TR_018 — Normalisation dropdown: Z-score, Min-Max, None options. *TR_019 — SMOTE and Class weights options visible when imbalance present.*

US-019 — SMOTE / Class Imbalance Control

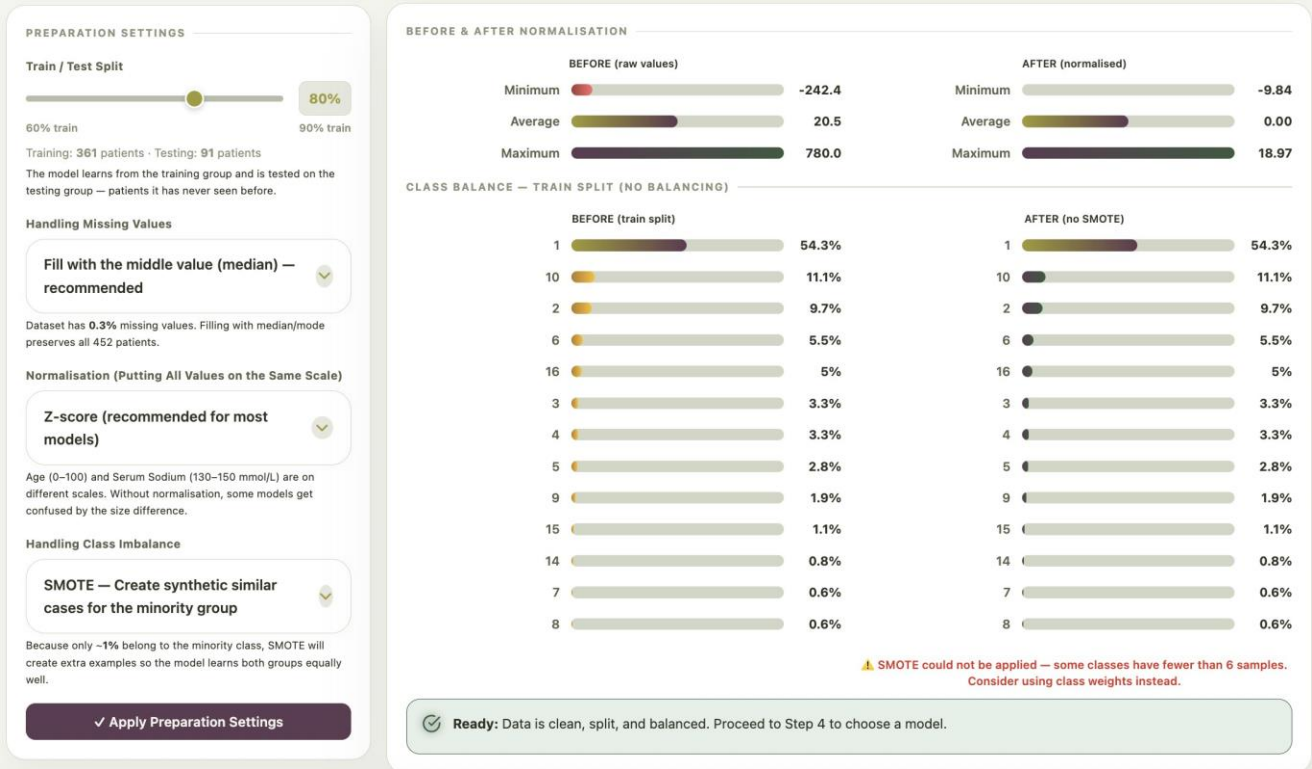
TR_019 shows the 'Handling Class Imbalance' section with SMOTE selected and 'Class weights' as an alternative. TR_020 shows the result after applying SMOTE: the minority class is balanced to 50% / 50%.



TR_020 — Before/After SMOTE: ckd 61.9% → 50% / notckd 38.1% → 50%. Also shows Before/After Normalisation comparison.

US-020 — Apply Preparation Settings & Before/After Comparison

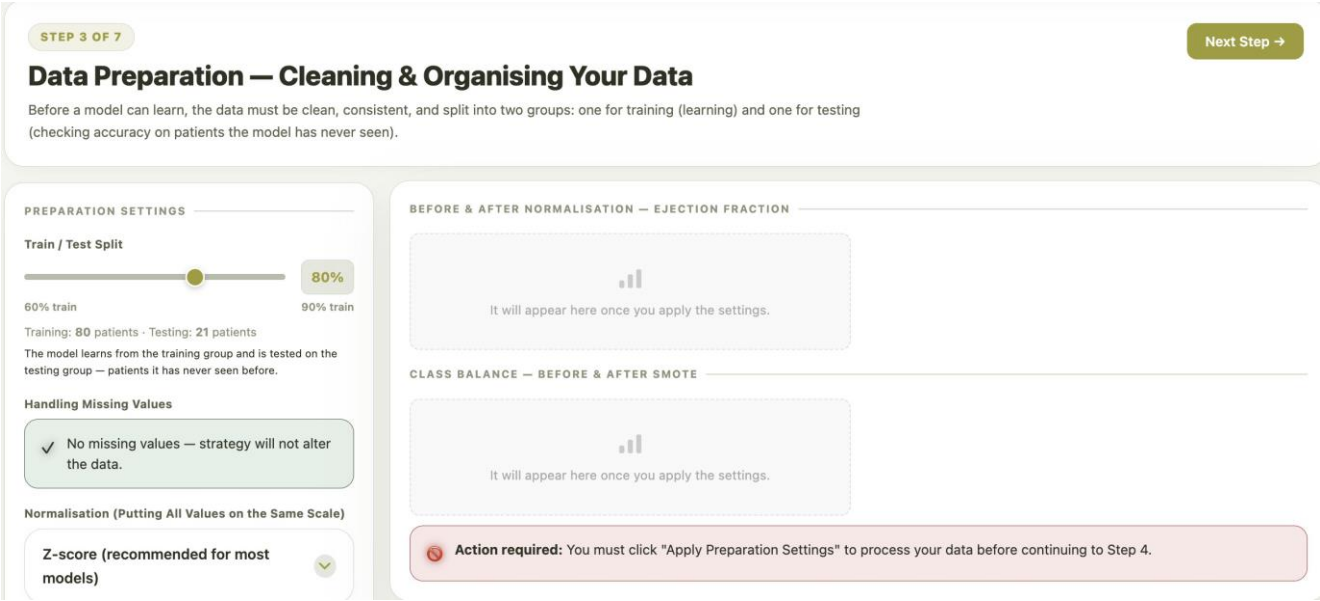
TR_021 shows the full Step 3 panel after clicking 'Apply Preparation Settings'. The right panel displays the Before/After Normalisation chart (Min: -242.4 → -9.84, Average: 20.5 → 0.00, Max: 780.0 → 18.97) and the Class Balance before/after chart. A red warning message indicates that "SMOTE could not be applied — some classes have fewer than 6 samples. Consider using class weights instead." A green 'Ready' banner confirms the pipeline is prepared for Step 4.



TR_021 — Full Step 3 applied: Z-score normalisation + SMOTE settings, before/after charts rendered, green Ready banner.

US-021 — Gate Control: Step 4 Blocked Until Preparation Applied

TR_016 confirms the Step 4 gate: before clicking Apply, the right panel shows placeholder charts with the message 'It will appear here once you apply the settings' and a red 'Action required: You must click Apply Preparation Settings to process your data before continuing to Step 4.' banner. Once applied (TR_020, TR_021), the green 'Ready: Data is clean, split, and balanced. Proceed to Step 4.' banner replaces it.



TR_016 — Step 4 gate active: empty chart placeholders and red action required banner.

Step 3 — Test Result Summary

User Story	Test Description	Result	Evidence / Screenshot
US-016	Train/test split slider — default 80%, configurable, patient counts shown	**PASS**	TR_016, TR_017
US-017	Missing value handling: Median / Mode / Remove — all selectable	**PASS**	TR_017
US-018	Normalisation: Z-score / Min-Max / None — all selectable	**PASS**	TR_018
US-019	SMOTE class imbalance control visible and selectable when relevant	**PASS**	TR_019, TR_020
US-020	Apply settings: before/after normalisation and SMOTE charts displayed	**PASS**	TR_020, TR_021
US-021	Step 4 gate: blocked before apply; green ready banner after apply	**PASS**	TR_016, TR_020, TR_021

Full Test Execution Summary — All User Stories

User Story	Test Description	Result	Evidence / Screenshot
US-001	Select specialty from domain pill bar	**PASS**	TR_001, TR_003, TR_015
US-002	Confirmation dialog on domain switch	**PASS**	TR_002, TR_033
US-003	Privacy notice on all screens	**PASS**	TR-023
US-004	Step 1 shows clinical context content	**PASS**	TR_001, TR_003
US-005	Use built-in example dataset	**PASS**	TR_008, TR_009, TR_010
US-006	Upload valid CSV — 5 files accepted	**PASS**	TR_024–TR_028
US-007	Invalid files rejected with error messages	**PASS**	TR-023, TR_029–TR_032
US-008	File exceeds 50 MB rejected	**PASS**	TR_032
US-009	Fewer than 10 rows rejected	**PASS**	TR_029
US-010	No numeric column rejected	**PASS**	TR-023
US-011	Column summary with missing value indicators	**PASS**	TR_009
US-012	Outcome distribution displayed	**PASS**	TR_008, TR_015
US-013	Class imbalance warning shown when relevant	**PASS**	TR_015
US-014	Column Mapper: target/feature assignment flow	**PASS**	TR_006, TR_007
US-015	Step 3 gate: blocked before save / open after save	**PASS**	TR_004, TR_005, TR_011
US-016	Train/test split slider functional	**PASS**	TR_016, TR_017
US-017	Missing value handling dropdown — 3 options	**PASS**	TR_017
US-018	Normalisation dropdown — 3 options	**PASS**	TR_018
US-019	SMOTE control visible and selectable	**PASS**	TR_019, TR_020
US-020	Apply settings: before/after comparison shown	**PASS**	TR_020, TR_021
US-021	Step 4 gate: blocked before apply / ready after	**PASS**	TR_016, TR_020

Overall Verdict: ALL 21 USER STORIES PASS
32 screenshots demonstrate full functional coverage of Steps 1, 2, and 3. No defects identified. All acceptance criteria verified.